

Ramratan Padhy

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EDUCATION

SRM University AP

Bachelor of Technology in Computer Science Engineering

- Year 1 | SGPA: 9.0/10.0

Amaravati, Andhra Pradesh

2025 – Present

EXPERIENCE

Next Tech Lab AP

Associate

Amaravati, Andhra Pradesh

Oct 2025 – Present

- Research associate at a QS ReImagine awarded student society.
- Collaborating with a research collective to explore low-level implementations of emerging technologies.
- Contributing to peer-to-peer technical workshops on systems programming and AI architecture.

Notecove (Stealth Startup)

Technical Content Writer

Remote

May 2025 – Aug 2025

- Authored 30+ comprehensive technical articles and guides on emerging software trends and AI ethics.
- Synthesized complex engineering concepts into accessible documentation for a specialized audience.
- Collaborated with the founding team to define the technical voice and content strategy for the platform.

Hush (Startup)

ML Intern

Remote

March 2026 – Present

- Worked with image models to discern user profiles to keep the app authenticity.
- Collaborated with the founding team to define and enhance user experience.

PROJECTS

AI Advertisement Tool (RAG Pipeline) | Python, RAG, ETL

Completed

- Developed an end-to-end content engine using a Retrieval-Augmented Generation (RAG) framework to synthesize data from LinkedIn, GitHub, and web portfolios.
- Architected an ETL pipeline to clean unstructured web data for LLM-based script generation and automated video/audio synthesis.

High-Performance Editorial Portfolio | HTML, CSS, Web Optimization

Completed

- Architected a zero-dependency, single-file portfolio site optimized for instant edge delivery, achieving sub-300ms load times and a perfect 100/100 Lighthouse performance score.
- Engineered a completely static, JavaScript-free architecture that inherently eliminates third-party vulnerabilities, maximizing security while ensuring optimal crawlability for modern AEO and SEO bots.

Temporal RAG pipeline | Python, RAG, ETL

Completed

- Developed an end-to-end pipeline where you can input a video, and make any query regarding it.
- Worked with advanced data structures and various audio-video technologies to produce a system capable of answers queries about a video based on timeline and not only semantical similarity.

Anti-UAS (Unmanned Aerial System) Implementation | Python, Physics Modeling

Completed

- Simulated anti-drone defense logic focusing on the mathematical modeling of projectile trajectories and signal interference using Newtonian physics.

Game Development Framework | Python, Godot, Networking

Completed

- Shipped a playable AI debate game (Ace Attorney-style) where an LLM opponent argues back and a separate judge LLM scores logical soundness. Built on Godot, Python, and Groq with a RAG layer for character personality.

Independent Research on BitNet-1.58b | Python, Machine Learning, Mathematics

Ongoing

- Conducting novel research and experimentation on BitNet-1.58b by altering its context length.

Independent Research on SWA-based LLMs | Python, Machine Learning, Mathematics

Ongoing

- Conducting novel research and experimentation on Qwen 2.5 family by attempting to quantize it using BitNet's ternary engine.

TECHNICAL SKILLS

Languages: Python, C, C++ (Basic), LaTeX,

Developer Tools: Git, VS Code, Postman, APIs, Vector Databases

Core Concepts: Data Structures, Transformer Architectures, RAG, Technical Documentation, Mechanistic Interpretability

Soft Skills: Technical Communication, Research Synthesis, Problem Solving, Teamwork